

XOR the String

A binary string is a string consisting of 0s and 1s. You are given a binary string $s = s_1 s_2 \dots s_n$ of length n and an integer $k \geq 0$.

For a binary string $t = t_1 t_2 \dots t_m$ ($m \geq 2$), **transforming** t means changing t to $t_1 (t_1 \oplus t_2) t_2 (t_2 \oplus t_3) t_3 \dots t_{m-1} (t_{m-1} \oplus t_m) t_m$

where $\oplus$ is the bitwise XOR operator. For example, 011 transformed is 01101, and 10011 transformed is 110001101. Here is an image showing how 10011 is transformed:

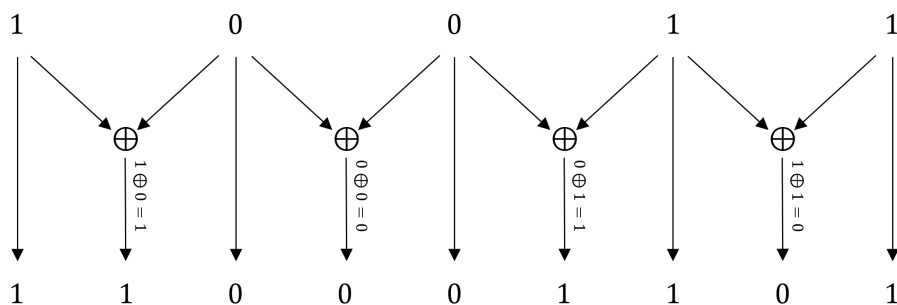


Figure 2: Illustration of how 10011 is transformed into 110001101. Between two characters, the XOR of the characters is inserted between them.

The **beauty** of a binary string $t = t_1 t_2 \dots t_m$ ($m \geq 2$) is the number of integers i ($1 \leq i < m$) such that $t_i = t_{i+1}$. We will denote the beauty of t by $f(t)$. For example, $f(10010111) = 3$, $f(010) = 0$, and $f(000) = 2$. Also, define $g(t)$ to be the value of $f(t')$, where t' is t transformed k times.

Output the sum of $g(t)$ over all substrings of s with length at least 2. In other words, let $s[l, r]$ be the substring $s_l s_{l+1} \dots s_r$. Output $\sum_{1 \leq l < r \leq n} g(s[l, r])$. Since the answer might be very large, output the answer modulo 998244353.

Input Format

The first line contains two space-separated integers, n and k ($2 \leq n \leq 2 \cdot 10^5$, $0 \leq k \leq 10^{18}$).

The second line contains s , a binary string of length n .

Output Format

Output a single integer, the sum of $g(t)$ of all substrings t of s with length at least 2, modulo 998244353.

Constraints

Task	1, 2	3	4	5	6	7	8	9
n	(samples)	5000	15	200000	500	5000	200000	200000
k	(samples)	0	10	0	500	10^{18}	10^6	10^{18}
Points	0	5	10	10	20	20	15	20

Sample Input 1

3 2
101

Sample Output 1

4

Sample 1 Explanation

There are three substrings of length at least 2: 10, 01, and 101. The substrings are transformed $k = 2$ times as follows:

- $10 \rightarrow 110 \rightarrow 10110$ (beauty 1)
- $01 \rightarrow 011 \rightarrow 01101$ (beauty 1)
- $101 \rightarrow 11011 \rightarrow 101101101$ (beauty 2)

The answer is $1 + 1 + 2 = 4$.

Sample Input 2

2 30
00

Sample Output 2

75497471

Sample 2 Explanation

Remember to output the answer modulo 998244353.